Pest Management for Vegetable Bedding Plants

Introduction

Vegetable bedding plants and/or transplants are commonly grown in New England in greenhouses for field setting or as part of the spring sales mix for resale at farm market stands. Many vegetable crops are grown from transplants in New England due to the late spring, short growing season and desire to obtain mature, harvestable crops as soon as possible.

Transplants can be grown in greenhouses under cover to provide a protected environment. One of the most critical features is a source of heat to provide appropriate temperatures. A frequent question by growers is the use of supplemental heaters in the spring. When growing transplants in the greenhouse or high tunnel, be careful not to use unvented heaters. An unvented heater is one that is designed without a flue connection so that the heat and products of combustion are exhausted into the greenhouse. Unvented heaters can be fired with natural gas, propane or kerosene which all contains traces of sulfur. During combustion sulfur in the fuel is combined with oxygen to form sulfur dioxide. Levels as low as 0.5 part per million (ppm) can cause injury to some plants. Once the sulfur dioxide enters the plant through the stomates, it reacts with water to produce sulfuric acid that causes leaf burn, flecking and general chlorosis. Tomatoes and white petunias are very sensitive and will show signs in as little as one hour. Ethylene gas is another pollutant formed during combustion. Ethylene levels as low as 0.01 ppm can cause symptoms such as malformed leaves and flowers, stunted growth, bud abscission, epinasty and flower senescence.

Although vegetable bedding plants may only be in the greenhouse for a short period of time, it is important to produce a high quality pest-free transplant. Scheduling, plant nutrition, greenhouse management (see additional references at the end of this section) and pest management all influence the quality of transplants.

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Types and Varieties

For Field Production: Consult with your seed supplier and review the individual crop sections in the manual for suggested varieties to grow for field production. Grow the crops at appropriate temperatures. Pay particular attention to scheduling times, light, temperature and nutritional

requirements needed to grow healthy transplants.

With the exception of a few perennial vegetables, vegetable plants are started from seed. Brussels sprouts, broccoli, cabbage, lettuce and tomatoes are easy to transplant vegetables that are able to absorb water efficiently and form new roots rapidly. Vegetable plants that are a little more difficult to transplant, do not absorb water as efficiently, but form new roots quickly include cauliflower, eggplant, onion and pepper. Vegetable plants that are difficult to transplant include cucumbers, melons, squash and sweet corn.

For Spring Bedding Plant Sales: There are so many choices, from gourmet greens and vegetable amaranth (popular in Southern Asia, Africa and West Indies) to yellow cherry tomatoes and an assortment of colored peppers and eggplants. To find new varieties to grow for spring bedding plant sales, see the All American Selection (AAS) Winners website, the National Garden Bureau website and your favorite seed supply company catalogues. State University trial results can also help you select varieties that will perform best in your area. Create “edible containers”, by combining flowers and vegetables in mixed planters. Swiss chard, ‘Bright Lights’ with pretty colored stems in reds, pink, yellow and orange are easily grown and are great for containers or in the garden.

Vegetable Bedding Plant and Transplant Fertility

There are numerous factors affecting the growth of vegetable bedding plants and transplants. Factors which have a dramatic effect on growth are the types of growing media, watering practices and fertilization programs used in the greenhouse.

Types of Growing Media

Growing media for vegetable bedding plants or field transplants in greenhouses contain a variety of soilless ingredients such as peat moss, vermiculite, perlite, shredded coconut husks (coir), or composted materials plus starter nutrients and a wetting agent. Field soils are generally unsatisfactory for the production of plants in containers because soils do not provide the aeration, drainage and water holding capacity required. They also need to be pasteurized or fumigated to prevent the development of diseases and germination of weed seeds.

Growing media are designed to achieve high porosity and water retention while providing adequate aeration. Premixed media is common in the greenhouse industry. Suppliers offer a diversity of mixes that are available prepacked (in bags, bales, super sacks) or in bulk. Recipes are specially formulated for propagation, specific crops or general greenhouse crops. Soilless media purchased in bags does not have to be pasteurized or fumigated before use. Preventative applications of biological fungicides or fungicides may be necessary with some vegetables that are prone to damping off. Growers can obtain commercially available mixes with different types of biological fungicides added to the mix.

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Compost-based mixes are also available commercially as a substitute for traditional soilless media, especially for organic production. See section on organic nutrient management.

Steam treatment

Field soil is not recommended for vegetable transplants in greenhouses. However, if it is used, it must be treated to eliminate soilborne plant pathogens, insects, and weed seeds. Once the soil has been treated, care must be taken to avoid reinfestation.

Treatment of soil with steam is preferred over chemicals because it is very effective and safe. All pathogens will be killed by steam and only a few of the hardiest weed seeds will survive. There are portable steam generators available.

Ideally, the temperature of the entire soil mass should be raised to 160° to 180°F for 30 minutes. It is important to use several accurate thermometers placed in one or more corners and in the center of the soil. Since it is difficult to avoid the soil temperature rising to 212°F, bring it up to this level and then reduce the pressure to allow the temperature to drop to 180°F in the coldest part of the soil mass.

The soil moisture content prior to steaming is important. Overly moist soil will take a long time to reach proper temperature. However, some moisture is necessary for the effective killing of microorganisms as well as the conductance of heat. Proper soil moisture for steaming is approximately the same for good planting conditions; soil squeezed in the hand should crumble easily. If possible, the soil mass should be moistened evenly two to three weeks prior to treatment. This will germinate difficult-to-control weed seeds, such as oxalis and clover, making them susceptible to heat. Prolonged steaming of soil at temperatures higher than 180°F can result in undesirable side effects such as overkill of beneficial soil microflora and accumulation of ammonium and manganese. Soil high in organic matter should be tested for ammonium after steaming. Several weeks may be necessary to allow for the dissipation or conversion of the ammonium. The incorporation of dolomitic lime and superphosphate, based on a soil test, may reduce ammonium levels. High levels of ammonium can reduce and even stall plant growth.

Plant Nutrition

pH: The optimum pH range for vegetable bedding plants grown in soilless media is 5.5 to 6.5. Limestone in the growing media, irrigation water pH and alkalinity, and the acid/basic nature of fertilizer solution used are major influences on the pH. Have your soil tested each month to adjust your fertilizer program to manage the pH of the growing media and to prevent problems.

Water Quality: The quality of water used for irrigation affects plant growth. Water containing a large concentration of dissolved salts can cause excess soluble salts damage. Water with a high alkalinity can caused plant damage from carbonates and bicarbonates through increased root medium pH over time. It is advisable to have your water tested prior to the spring growing season. Your greenhouse fertilizer program should be adjusted according to test results.

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Fertilizer Injectors: In conventional greenhouses, nutrients are delivered using various water-soluble fertilizers through a fertilizer injector, through the use of controlled-release fertilizers, or using a combination of the two.

Fertilizer injectors are used in liquid feeding systems. These devices inject a small quantity of concentrated fertilizer solution (stock solution) into the irrigation line so that the water leaving the hose (dilute solution) supplies the proper concentration of fertilizer. Rates of fertilization are often

given in parts per million (ppm) of nitrogen, which is a way of expressing the fertilizer concentration. The amount of fertilizer to dissolve per gallon of water (stock solution) to make the appropriate concentrate for a specific injector setting needs to be determined. This information is listed on the bag of fertilizer. An injector setting of 1:100 indicates that 1 gallon of fertilizer concentrate delivers 100 gallons of final solution and is not an indication that the injector is delivering 100 ppm.

Choosing Fertilizers: Factors to be considered when choosing fertilizers include the ratio of ammonium to nitrate-N, trace element starter charge, content of calcium and magnesium, and potential acidity or basicity. Commonly used fertilizers include 15-0-15 (Dark Weather Feed), 15-15-15, 15-16-17 and 20-10 20 or Cal-Mag 15-5-15.

Peat-Lite Specials (15-16-17, 20-10-20). These fertilizers are among the most popular for routine fertilization of vegetable bedding plants. Both are high (>50%) nitrate fertilizers. However, these fertilizers also have elevated trace element levels which may raise iron (Fe) and manganese (Mn) to toxic levels at low pH. Both are acid-forming fertilizers, but 20-10-20 has the greater potential acidity.

15-15-15 Geranium Special. "Triple 15" is a good alternative to the Peat-Lite Specials for crops sensitive to trace element toxicities. Trace element levels supplied by this fertilizer are lower than the Peat-Lite Specials. Otherwise, at the same rate of nitrogen (N), plant response will be very similar to 15-16-17. This is an acid-forming fertilizer also; the potential acidity is slightly greater than 15-16-17.

20-20-20 General Purpose. Growers who use this fertilizer on soilless media risk ammonium toxicity because the nitrogen in this fertilizer is 75% ammonium and urea. Some growers who use media containing soil do not appear to have problems. If 20-20-20 is used, the growing medium should be tested frequently for ammonium. 20-20-20 supplies trace elements and has the greatest potential acidity of fertilizers commonly used in New England greenhouses. Tomato, eggplant and pepper plants are especially sensitive to ammonium. High ammonium levels, especially in soilless mixes, can reduce plant growth and cause yellowing of the foliage.

Low Phosphorus (P) Fertilizers (20-0-20, 20-1-20, 15-0-15). These fertilizers can be tried as an alternative to chemical growth regulators for vegetable bedding plants. This technique of growth control is sometimes called "phosphorus starvation." It is generally believed that more P than necessary is being applied to greenhouse crops. Too much P may cause plants to stretch and P is a ground water pollutant. Unfortunately, in terms of height control, these fertilizers may be of no benefit if they are applied to a growth medium containing superphosphate or a high starter charge of P. Also, there is a risk of P deficiency if the fertilizers are used continuously with low P growth media. The low P fertilizers are quite different in many ways. 15-0-15 and 20-0-20 supply Calcium (Ca). 15-0-15 is a basic (raises pH) fertilizer containing about 95% nitrate and 20-0-20 is a neutral fertilizer. 20-1-20 is an acidic fertilizer and it does not supply Ca, but it is about 70% nitrate.

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Calcium nitrate and potassium nitrate (15-0-15). Use of this fertilizer combination greatly reduces the chance of trace element toxicities. Some growers alternate its use with the Peat-Lite Specials on a 2-3 week basis to supply Ca and to counter the acidic effect of the Peat-Lites. However, both superphosphate and a trace element fertilizer must be incorporated in the growing medium if this combination is to be used as the sole fertilizer.

Nitrogen, Phosphorus, Potassium

Nitrogen. Nitrogen concentration in the greenhouse fertilizer program has a greater affect on the growth of transplants in the greenhouse than either phosphorus or potassium. Increasing the level of nitrogen results in taller transplants with thicker stem diameters and heavier plant weights. Applying too much nitrogen in the greenhouse results in soft, poor quality transplants. These lush transplants may also be more prone to phloem feeding insects such as aphids and to foliar blights.

Phosphorus: Phosphorus has a limited affect on the growth of bedding plants when compared to nitrogen, but should be included as part of a complete fertilizer. Increasing the phosphorus concentration results in a moderate increase in transplant height, stem diameter, and shoot fresh and dry weight. If phosphorus is restricted to the point at which the plants show extreme phosphorus deficiency (purple leaves and stems, stunted plants), field performance will be reduced.

Potassium: Potassium has the least affect on the growth of plug tomato transplants of the three major nutrients. Adequate potassium is applied as part of a complete fertilizer.

Guidelines for Rates and Frequency of Fertilizer

Small, slow-growing plants should receive lower rates or less frequent application until they are well-established. Care should be taken not to over-fertilize vegetable bedding plants to avoid overgrown plants. Young seedlings are especially vulnerable to injury from high soluble salts.

While plants are in the plug or seedling stage, use a complete water soluble fertilizer at the rate of 50 – 100 ppm N every time plants are watered and use clear water (no fertilizer) every third watering. Use the lower rate (50 ppm) early and the higher rate (100 ppm) later if the seedlings are to be held in the flat or tray three or more weeks before transplanting. Shortly after transplanting, as plants approach rapid growth, increase the rate to 200 ppm N at every watering or 300 ppm N once every 7 days, watering with clear water 2 or 3 times in-between each fertilization.

Fertilizer Solution Volume: The volume of fertilizer solution applied has a dramatic affect on the growth of the vegetable bedding plants. As the volume of water-soluble fertilizer increases, the quantity of nutrients delivered to the plant also increases resulting in an increase in height, stem diameter and plant weight. Doubling the volume applied also doubles the amount of each nutrient potentially available to the plant.

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Plant Growth Rate and Environmental Conditions. In general, nutrient requirements of vegetable bedding plants are greatest during periods of rapid growth. Too much fertilizer during slow growth periods may lead to excess soluble salts; failure to provide enough fertilizer during

periods of rapid growth will lead to nutrient deficiency.

Nutritional Problems

Vegetable bedding plants and transplants are subject to the same nutrient disorders as other plants. Early in production serious problems are: high soluble salts, trace element toxicities, and ammonium toxicity. Late in production, particularly in cell packs, plants may develop nitrogen deficiency symptoms as the earliest indication of insufficient fertility.

Soluble Salts. Injury to vegetable bedding plants from excess salts seems to be most common shortly after transplanting. Seedlings are much less tolerant than established, rapidly growing plants. Some soilless mixes may contain enough “starter charge” to cause excess salts problems in the first few weeks after transplanting, particularly when a water-soluble fertilizer is also applied. Excessive drying, poor drainage, and uneven watering are factors which can aggravate this problem. Check roots of plants often and conduct regular soil tests to identify and prevent problems. It is difficult to diagnose a soluble salts problem by symptoms alone. Often nutrient deficiencies and root diseases cause the same symptoms. Therefore, a greenhouse (not field) soil media test is advisable.

Trace element toxicities. Iron (Fe) and/or manganese (Mn) can be accumulated to toxic levels by tomato plants. Symptoms appear as numerous small dark spots and mottling of the foliage. The potential sources of excess Fe and Mn are: trace element fertilizers in the mix, water-soluble fertilizers with elevated trace elements levels, and sometimes irrigation water. Low growth medium pH aggravates the problem by increasing Fe and Mn availability. Toxicity can be avoided by keeping the pH in the range of 5.8 – 6.0 for susceptible crops and by the use of fertilizers with lower trace element levels.

Ammonium toxicity. This is less common today because most growers use water-soluble fertilizers that supply about 50/50 ammonium and nitrate to fertilize plants in soilless media. Tomato, eggplant, and pepper are most sensitive to ammonium nitrogen, but many other vegetable bedding plants can be harmed if ammonium becomes excessive. Too much ammonium during the early spring (February or March) in low light and cool media conditions can be toxic to plants. Plant growth may be reduced with yellowing of the foliage.

Organic Nutrient Management

The quality of the planting mix is important to insure proper plant health in organic production. Conventional growing media that contains synthetic ingredients (wetting agent, chemical fertilizer) cannot be used in organic production of field transplants, and vegetable bedding plants. However, acceptable growing media can be composed from a wide variety of approved materials. These organic blends may be purchased off-the-shelf, custom-blended by manufacturers, or produced on the farm.

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Most commercial potting mixes contain synthetic fertilizers and wetting agents that do not meet organic standards. One alternative is to arrange a special order from a commercial supplier who agrees to exclude starter fertilizers and wetting agents and then, plan to add your own. Purchasing a commercially prepared organic mix is the easiest way to get started and most

growers choose this option to reduce the risk of soil-borne diseases. Common components such as peat moss, perlite, vermiculite, and coconut coir are generally acceptable for organic certification, but check with your organic certifier. Compost, being the most renewable, is a preferred material for many organic growers.

Dr. John Biernbaum, from Michigan State University, for his research, chooses a 50/50 mixture of peat and compost with a pH of 6.0 as an organic potting media. Dr. Biernbaum makes his own compost from garden waste, straw, hay and sheep/horse manure and screens the compost to provide a uniform product. This is just one of many options; however the finished compost product needs to have good physical, chemical and biological properties.

Since compost can vary from batch to batch, there can be variation in the performance of mixes, even when using the same recipe. If making or buying compost for your potting mix, try to have it ready at least 6 months before you need it. This allows time for the compost to mature so that nutrients are stabilized, phytotoxic compounds have degraded and disease suppressive beneficial microbes have a chance to increase. It is very important to test the compost before you use it to determine pH, available nutrients, and soluble salt levels. Organic potting mixes may contain from 20 to 50% compost by volume depending upon the crop, container size and growing conditions. 100% compost is generally not advisable.

Another option recommended by Eliot Coleman, from Four Season Farm, Maine is to use a blend of peat and compost or peat-based soilless medium and compost mixed with 14 lbs. per cubic yard of equal parts blood meal or alfalfa meal for nitrogen, rock phosphate for phosphorus and green sand or organic approved potassium sulfate for potassium. He lets the blend sit for a month or more before use.

A third option is to use mature, well-balanced compost blended with peat and possibly perlite and/or vermiculite for aeration to supply all the nutrients to grow and finish transplants. Nutrient sources such as alfalfa, alfalfa meal and other organic approved components can be incorporated during the composting process. Compost will mature during the fall and can be stored for use until spring. These last two options add some nutrients to the root medium, so plant nutrition is not dependent on liquid fertilizer. However, these options require purchasing the organic nutrient sources and making sure the rate and method of application are correct.

More information on organic growing media can be found in the publication, "Potting Mixes for Certified Organic Production □"

There are also many fertility management options such as supplementing with liquid organic fertilizers. Many growers are familiar with fish fertilizers made from waste products of the ocean fish processing industry. The material is a thick, heavy liquid supplying plant nutrients at presumably varying levels of availability. Fish fertilizers probably supply mostly ammonium nitrogen which could be a disadvantage for some plants. Also, fish fertilizers can be a problem to store diluted because they spoil and they can be difficult to inject through some systems. In our area the Neptune's Harvest brand is the most commonly available fish fertilizer and it is OMRI-approved for organic greenhouses.

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In his studies, Dr. Biernham, used liquid fish emulsion (5-1-1) as a sole fertilizer with a soilless peat based medium and coconut coir successfully for several months. Fish emulsion fertilizers are likely to have an odor and fertilizing less often may be preferred, for example, every two weeks or once a month. The rate applied will vary depending upon how often one fertilizes. Plants fertilized with organic fertilizers will not show the rapid growth response seen with synthetic fertilizers. With this method, Dr. Biernbaum suggests beginning with a growing media containing 60-70% peat and 30-40% perlite and/or vermiculite without fertilizer and wetting agent, but limed to a pH of 5.5 to 6.5. Apply the water soluble organic fertilizer as needed, usually soon after seed emergence or at transplanting.

Many growers are using Daniels 10-4-3 a liquid, “organically-based” fertilizer. The organic portion is oilseed extract. Most of the nutrients, however, are derived from inorganic salts and for this reason it cannot be certified as being organic.

Daniels Pinnacle 3-1-1 is a less well-known liquid fertilizer. It is an organic fertilizer, as most nutrients are derived from oilseed extract and extra nitrogen is provided by sodium nitrate (“Chilean nitrate”). This fertilizer is a great step forward in finding organic fertilizers that can be easily applied by growers using the systems to which they are accustomed. However, Pinnacle spoils after dilution and it does not provide adequate nutrition at the label rates with the result that growth is reduced and nutrient deficiency symptoms develop, likely related to high pH and iron deficiency. Pinnacle seems to work better when it is alternated with fish emulsion. There are no deficiency symptoms, pH is lower and plant nutrient status is better.

Planting

Handling Growing Media

How soilless growing media is handled can greatly influence the air space and available water for the plant roots. The major concern is to preserve the air space or porosity to insure healthy root growth. You want to prevent compaction that encourages damping off diseases and poor root growth. Containers, including plug trays, should be lightly filled and the excess media brushed off the top. At no time should any growing containers be stacked. Stacking containers causes compacted media. This damage cannot be remedied after the compaction has developed. Add water to peat-based mixes before filling plug trays to help create more aeration. If mixing your own media, thoroughly mix components, but do not over-mix.

Production Schedules

Starting seeds too soon, will result in overgrown plants of poor quality. The following are guidelines for growing vegetable bedding plants. Note the number of weeks from seed to sale or transplant. This will vary according to different growing conditions and should serve only as a guide.

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Crop	Germination Temperature (°F)	Optimum Day Production Temperature (°F)	Minimum Night Temperature (°F)	Weeks from Seed to Sale	Weeks from Seed to Transplant in Field
Basil	70 +	70 to 75	60 to 65	4 to 7	4 to 6
Broccoli	70 to 75	65 to 70	55 to 60	4 to 7	4 to 6
Cabbage	70 to 75	65 to 70	55 to 60	4 to 7	4 to 6
Cauliflower	70 to 75	65 to 70	55 to 60	4 to 7	4 to 6
Celery	70 to 75	60	60	5 to 7	8 to 10
Chard	70	60	50		
Cucumber	70 to 75	70 to 75	60 to 65	2 to 3	3
Eggplant	70 to 80	70 to 80	60	7 to 9	6 to 8
Lettuce	65 to 70	60 to 65	50	3 to 5	
Leeks	75	65 to 70	55 to 60	10 to 12	10 to 12
Melons	70 to 85	70 to 75	60 to 65	2 to 3	2 to 3
Onions	75	65 to 70	55 to 60	10 to 12	10 to 12
Peppers	75 to 85	70 to 75	60	6 to 8	6 to 8
Squash/ Pumpkin	70 to 85	70 to 75	65	2 to 3	2 to 3
Tomatoes	75	65 to 75	60	5 to 8	5 to 8
Watermelon	80 to 90	70 to 80	65 to 70	3 to 4	3 to 4

Reference: Tips on Growing Bedding Plants, Ohio Florists Association)

Note: The greater the difference between daytime and nighttime temperatures, the more plants will "stretch" (stems elongate). See section on Managing Plant Height.

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Germination Tips

Warm temperatures and uniform moisture are needed to ensure successful germination and get the plants off to an even start. Many germination chamber systems are commercially available including custom built germination units. Many growers use bottom heat or root zone heating to provide warm, even temperatures. Rubber tubing or mats with hot water are placed on the bench top under the plants. A weed mat barrier is placed on the top of the bench to help spread the heat with skirts on the side to help contain the heat. In all systems, it is important to remove flats from the germination chamber as soon as radicles break through the seed coat to avoid seedling stretching. Experience and experimentation with your total seeding system is the key to uniformity and success.

Celery

Celery seeds germinate best at 70 – 75°F. To prevent bolting, maintain greenhouse temperatures above 55°F

Cole Crops (Cabbage, Broccoli, Brussels sprouts, Cauliflower)

To prevent premature seeding or bolting, avoid exposing transplants to temperatures below 50°F for long periods (week or more). The cold temperatures cause the development of premature heads or “buttoning” in cauliflower and broccoli. Any stress or check in growth results in a “wirestem” and plants will not become well established in the field or garden resulting in reduced yields and performance.

Eggplant

Eggplant seed can be directly sowed into 50 cell trays to shorten the time needed to produce transplants by approximately one week. Germinate seed in flats at 70 to 75˚F. Eggplants are susceptible to chilling injury and should not be grown below 40°F. Any stress or check in growth will result in tough woody stems and transplants that will have a tough time getting started later in the field or garden.

Tomatoes

Tomato seeds germinate best at 75˚F. As soon as there is any evidence of germination, they should be removed from the mist and bottom heat. The ideal root-zone temperature is 77-86°F during the first four weeks of growth and 68 to 77°F during the fifth and sixth weeks. Optimal growing-on day temperatures are 65 to 75°F with minimum night temperatures of 60°F. Exposure of tomato plants to temperature below 60°F will likely result in rough fruit (catfacing) on the first few clusters. Transplant young seedling into 2” to 3” containers when they have two true leaves and grow on until planted in the field. For earliest production, some growers finish their transplants into 6” or larger containers.

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Peppers

Seeds germinate at 85 to 90˚F. Note that germination is very slow at lower temperatures. Seedlings develop well at 75˚F daytime and 65˚ F night temperatures. Seeds may be directly sown into 72-cell trays for early production. Peppers are prone to damping off diseases especially if the media is compacted. They are also susceptible to transplant shock.

Vine Crops

Cucurbits do not transplant well, and are best to sown in the final container. After germination, excess plants can be thinned.

Transplants

Transplants can be grown in all types and sizes of containers. Before sowing, one needs to decide whether germination and finishing will occur in the same container or whether seeds will be sown in one container followed by transplanting to a finishing container.

Germinating and growing in small plugs requires more attention to detail and is probably best done by local, specialty propagators. Direct sowing of transplants to be finished in 200 – 128 cell or larger trays may be more manageable for many growers. Before making a decision, consider your available labor, and amount of greenhouse space, and the cost and benefit of each production method.

Plug seedlings should be transplanted as soon as possible after they have reached finished size.

Purchased plugs: Purchase transplants from a reputable local supplier to minimize the potential of importing severe disease and insect problems that are common in other regions of the country. Open and unpack the boxes immediately upon arrival and check the physical condition of the plants. Inspect plants for root and foliar diseases and for insects and mites. Report any damage or discrepancies immediately to your supplier (most companies want to hear within 24 hours). Photographs are also helpful.

Place plant trays on benches and water thoroughly with plain water (no fertilizer); be sure that plugs on the edges of the trays are thoroughly watered. Plugs can dry out quickly due to the small volume of growing medium; check the trays 2 or 3 times daily for watering. After the initial watering, apply a general-purpose fertilizer (such as 20-10-20) at 50 to 60 ppm of nitrogen at every other watering. Allow plants to acclimate to the greenhouse conditions for 24 to 48 hours before transplanting.

Transplanting to a finishing container: Water the plug trays thoroughly 2 to 3 hours before transplanting; this aids in removing the plugs from the trays. Prepare your cell packs or pots by filling them with pre-moistened growing medium and pre-dibbled holes for the plugs. Lightly fill containers and brush off excess. To prevent compaction, do not pack down or stack (“nest”) filled flats.

Take special care during transplanting to handle plants gently and avoid planting too deeply. Stems of tender seedlings can be easily injured when workers grasp or “pinch” the stems too tightly. This often leads to stem cankers causing plants to wilt and die. Plant plugs at the same depth as the original plug. Sometimes plants may have elongated stems and it is tempting to “bury” the stem. Resist the temptation, except for more adaptable tomato plants. See information under the specific crops for additional information on transplant production and planting.

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Plant Culture

Proper Watering Practices Hand watering and overhead irrigation systems are the primary methods of watering vegetable bedding plants and transplants. The amount of water and frequency of watering will vary depending on container size, growing media, greenhouse ventilation and weather conditions. It is important to water thoroughly, to moisten the entire container, which will promote root growth to the bottom of the container. If this is not done, root growth will develop in the upper part of the container and plants will be more prone to drying and drought stress. Allow plants to dry down before watering, but do not let the plant wilt severely, as this will damage roots. Vegetable bedding plants should be watered thoroughly early enough in the day to allow foliage to dry before evening. If foliage remains wet overnight, foliar disease problems will develop.

Limit water leaching to 10 to 20% to limit nutrient runoff. Most commercial mixes contain a wetting agent which provides initial hydration and improves wettability of the mix. Older mixes (stored longer than 8 months) are harder to wet and the addition of a liquid wetting agent may be needed.

Managing Plant Height

A review of pesticide labels indicates that Sumagic (uniconazole) is the only growth regulator labeled for use on a limited group of vegetable transplants (tomato, pepper, ornamental pepper, eggplant, tomatillo, ground cherry and pepino). Apply Sumagic only as a foliar spray at a rate of 2-10 ppm. As with any plant growth regulator, it is recommended to test growth regulator treatments on small crop samples and starting with a low rate before full-scale implementation. The maximum cumulative amount of Sumagic applied must not exceed 10 ppm with coverage of 2 quarts per 100 sq. feet. This means that total amount used in sequential applications can only add up to 10 ppm spray (example, 1 application at 10 ppm or two applications at 5 ppm or 4 applications at 2.5ppm). The last spray must be no later than two weeks after the two to four leaf stage of development. Experiments have shown that sequential applications produce the best results and that the earlier that the plants receive the Sumagic spray, the greater effect it will have on the final height of the transplants.

Since very few growth regulators are registered for vegetable bedding plants, plant height is often managed by adjusting temperature, water and fertilizer levels, or by physically brushing the plants. Research has shown that mechanical stress reduces stem elongation and maintains plant height. For example, brushing transplants twice daily for 18 days using about 40 strokes back and forth with a cardboard tube suspended from an irrigation boom can result in as much as a 30% reduction in stem elongation. Growers have also successfully used a wand made of plastic plumbing pipe or a flat piece of polystyrene foam. Vegetable plants such as tomatoes, eggplants and watermelons have responded to this mechanical manipulation. Note that this technique has damaged some tender plant species such as peppers and could also enhance the spread of disease.

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The greater the difference between daytime and nighttime temperatures, the more plants will “stretch” (stems elongate).When the day temperature is very warm and the night temperature is cool or cold, plants will be taller. If the day and night temperature are both the same, plants will

be shorter than with warm days and cool nights. If the night temperature in the greenhouse is kept warmer than the day temperature by using heating at night and ventilation during the day, the plants will be even shorter. Keeping day temperatures cool (<60) rather than letting the greenhouse get very warm (>70F) will help keep transplants shorter. The relationship is referred to as DIF, or difference between day minus night temperatures.

Water stress is another tool growers can use to manage plant height. Maintaining plants on the dry side limits cell expansion and plant growth. This method requires close attention to avoid permanent damage such as leaf burn or even plant death. One technique is to irrigate the growing mix thoroughly and then allow it to dry to the point where plants wilt before irrigating thoroughly again. Growth is restricted during the period when the growing medium is very dry. Once watered, the plants rapidly resume growth. Experienced tomato growers have successfully used this technique.

Withholding nutrients can also be used to prevent stretching. Low phosphorus fertilization is especially effective for tomatoes. If carefully managed, a mild to moderate phosphorus (P) deficiency may result in a desirable reduction in growth with no foliar symptoms of P deficiency. See section on fertility for more information.

Note: For field production the goal will be to put a vegetative plant out and promote rapid vegetative growth that will produce the largest yields. Water and nutrients must be carefully managed to produce healthy transplants.

Acclimating or Hardening off Transplants

The transition from the greenhouse to the field involves changes in light, temperature and wind. Vegetable transplants benefit by a gradual “hardening” off period before they are transplanted into the field. Gradual exposure to outdoor growing conditions and reduced watering at the end of the growing period with some protection from wind and temperature but full exposure to light can increase the survival rate of transplants in the field. Three to six days are adequate to acclimate transplants.

Care must be taken to not “overharden” young transplants. Cool-season crops exposed to very low temperatures can result in bolting (in cabbage) or buttoning (in broccoli or cauliflower). Warm-season crops generally are hardened at temperatures higher than those of cool-season crops. Cold temperatures can set back warm-season crops and can induce disorders such as catfacing in tomatoes.

Integrated Pest Management

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There are only a limited number of insecticides and fungicides labeled for greenhouse-grown vegetable bedding plants. Integrated pest management (IPM) offers a practical way to effectively manage pests on vegetable bedding plants and transplants. Through the use of sound cultural practices, monitoring techniques, accurate problem identification, and timely implementation and evaluation of appropriate management strategies, growers can improve their production while minimizing their reliance on routine pesticide applications. IPM utilizes many different management options; genetic, cultural, physical, mechanical, biological and chemical. Routine

crop inspection alerts growers to developing pest and cultural problems while they are still minor and can be easily managed. Early detection and intervention is the foundation of an IPM program.

Disease Management

Diseases of vegetable bedding plants include *Botrytis* blight, damping-off, *Alternaria* blight, *Botrytis* blight, late blight, powdery mildew, downy mildew**,** bacterial diseases such as bacterial leaf spot, bacterial canker, and black rot, and viral diseases such as Cucumber Mosaic Virus (CMV), Tobacco Mosaic Virus (TMV), and Tospoviruses. Effective control of diseases requires accurate identification. Failure of disease control is often because the cause was not accurately identified. Symptoms caused by poor cultural practices can also mimic disease symptoms. Fungicides cannot correct problems caused by high soluble salts, poor aeration or a nutrient imbalance. An integrated approach to disease management involves the use of resistant cultivars, sanitation, sound cultural practices and the proper use of the correct pesticide.

Resistant Cultivars

Seed catalogues often feature disease resistant and tolerant varieties of vegetables. Utilize resistant varieties where feasible, but take some time to research the diseases that are giving you the most trouble to find other strategies to incorporate into the disease management plan.

Seed Treatments for Disease Management

Seed treatments are useful for many vegetable crops to prevent root diseases, as well as certain diseases carried on or within the seed. There are two general types of seed treatment: eradicative and protective. Eradicative seed treatments use hot water or chlorine to kill disease-causing agents on or within the seed. They are useful in controlling certain seed-borne bacterial diseases such as **bacterial leaf spot** on pepper and tomato and bacterial canker on tomato. Protective seed treatments use fungicides on the seed surface to protect the seed against decay and soil-borne organisms such as damping off caused by *Pythium, Phytophthora* and *Rhzoctonia*. For more information regarding seed treatments, contact your seed sales representative, Extension vegetable specialist or plant pathologist.

Sanitation

Pest management on vegetable bedding plants and transplants begins with a clean, weed-free, disinfected greenhouse. Before growing the crop, the greenhouse should be cleared of plant debris, weeds, flats and tools. Empty benches, potting tables, storage shelves, tools and cell packs should be washed and disinfected with a sanitizing agent. It is important to thoroughly clean or power wash to remove organic debris from plastic containers before using a sanitizing agent. Bits of organic debris can be difficult to remove and the organic matter can be a source of disease causing pathogens if the plug trays are reused.

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After the greenhouse has been sanitized, care must be taken to avoid recontamination with pathogens. Purchase certified, disease-free seed from reliable sources. If possible, purchase seed that has been disinfested by chemical and/or heat treatment by the seed company. Potting media is easily re-infested by dirty hose nozzles or tools and unsanitary growing conditions. The floor of the greenhouse is a good source for diseases. Use a hook to keep the hose nozzles off the floor. Grow transplants off the ground in a well-ventilated greenhouse. To prevent root rot diseases, avoid over-watering and over-fertilizing. Water early in the day to allow foliage to dry quickly to help prevent foliar diseases.

Use separate greenhouses for vegetable seedlings and ornamental bedding plants. Separate greenhouses: 1) will protect vegetable seedlings from any insect pests that may migrate from ornamentals and plants that are held over; 2) will protect vegetable seedlings from Tospoviruses; 3) protect curcurbit seedlings from powdery mildew originating on verbena and 4) facilitate treatment of the vegetable seedlings if pesticides are needed.

Keep tomato transplant production separate from greenhouse tomato fruit production. Greenhouses with both young transplants and mature plants increase the risk of perpetuating diseases.

Techniques to reduce high humidity

High relative humidity is one of the major contributing factors to *Botrytis* blight and powdery mildew, common fungal diseases of bedding plants. Warm air holds more moisture than cool air. During warm days, the greenhouse air is more humid. As the air cools in the evening, the moisture-holding capacity drops until the dew point is reached. Water then begins to condense on surfaces. Humidity can be reduced by exhausting the moist air and replacing it with cooler outside air that is drier. The method and time it takes to heat and vent depend upon the heating and ventilation system in the greenhouse. In greenhouses with vents, turn the heat on and crack the vents open about one inch. The moist humid air escapes from the vents. In greenhouses with fans, activate the exhaust fans for a few minutes and then heat the greenhouse to raise the air temperature. Then, shut off the fans. A clock can be set to activate the fans. The cooler, outside air will lower humidity levels as it is warmed in the greenhouse. A relay may be needed to lock out the furnace or boiler until the fan shuts off so that flue gases are not drawn back into the greenhouse. This will also help to prevent air pollution damage (ethylene or sulfur dioxide) to sensitive seedlings. Heat and vent two or three times per hour in the evening after the sun goes down and early in the morning at sunrise. Heating and venting can be effective even if it is cool and raining outside.

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Air movement, even in a closed greenhouse, helps reduce moisture on the plant surfaces and surrounding the plants. Using horizontal airflow (HAF) can also reduce condensation. HAF fans keep the air moving in the greenhouse, helping to minimize temperature differentials and cold spots. When the air is moving, it is continually mixed. The mixed air along the surface does not cool below the dewpoint so does not condense on plant surfaces.

In addition, cultural practices can be used to reduce humidity within the plant canopy. These include proper watering practices and spacing of plants. Since most vegetable bedding plants are grown in flats that are spaced flat to flat, reducing humidity within the canopy is difficult. Proper planting dates, plant nutrition, watering practices and height management techniques help to prevent lush, overgrown plants thereby reducing humidity within the canopy.

Always water in the morning to reduce the length of time the leaves stay wet after irrigating to prevent foliar diseases. Rising temperatures during the day will evaporate water from the foliage, so the leaves stay dry. Avoid watering late in the day or when water will sit on leaf surfaces for long periods of time.

Fungicides

Fungicides can provide excellent management of some diseases, but for others they may be ineffective. In general, to control root diseases, broad-spectrum fungicides should be applied as a drench on a preventative basis. Read directions for application on pesticide labels. An application of additional water may be necessary. For foliage diseases, obtain thorough spray coverage and treat when disease is first evident. provides a listing of fungicides labeled for vegetable bedding plants.

Biofungicides

Biofungicides are biological fungicides that contain living organisms such as fungi, bacteria, or Actinomycetes (a group of bacteria that form branching filaments) that attack plant pathogens and the diseases they cause. They can be used as part of an integrated disease management program to reduce the risk of pathogens developing resistance to traditional fungicides. Currently, there are no pathogens that are resistant to biological fungicides.

Biological fungicides may suppress diseases in a number of different ways. They may directly compete with the pathogen. The biological fungicide “shields” the roots by growing a defensive barrier around the roots. The microorganisms may produce an antibiotic or another toxin that kills the target organism. They may attack and feed upon the pathogen (mycoparasitism). As such, the biological fungicide must be present at the same time or before the pathogen appears. Some biological fungicides induce the plant to turn on their own defense mechanisms. Some of the advantages of using biological fungicides include: lower re-entry interval (REI) than many traditional fungicides, may be organic products (OMRI listed), may be less phytotoxic, and many can be used in rotation with synthetic chemicals. (See company web sites for more information on compatibility).

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Biofungicides should be used as a preventative treatment in conjunction with a regular monitoring program where root health and crop quality is evaluated. They will not cure diseased plants and must be applied before the onset of the disease. Biological fungicides need to be used in conjunction with standard cultural practices to help prevent diseases. Storage conditions, soil and air temperatures, and use of other chemicals affect their efficacy. Most biological fungicides also have a limited shelf life of one year. A number of products are commercially

available for use on vegetable bedding plants and transplants, please refer to the label for specific information regarding what specific types of diseases and specific plants are listed for each individual biological fungicide.

Specific Diseases

Blights

Botrytis blight

Botrytis can cause leaf blight, stem cankers, damping off and root rot. Plants may be attacked at any stage, but the new tender growth, freshly injured tissues and dead tissues are most susceptible.

Symptoms: *Botrytis* blight produces characteristic gray fuzzy appearing spores on the surface of infected tissues. Tan stem cankers can develop on basil.

Air currents and splashing water can easily disseminate the spores. In general, germination of spores and infection is dependent on a film of moisture for 8 to 12 hours, relative humidity of 93% or greater and temperatures between 55° and 65°F. After infection, colonization of plant tissues can occur at temperatures up to 70°F.

Management: *Botrytis* diseases can only be managed by a combination of methods including manipulation of environmental conditions (temperature, humidity and duration of leaf wetness), sound cultural practices and use of fungicides. Fungicides alone cannot control *Botrytis* and this pathogen has a long history of fungicide resistance development

- Control weeds and remove plant debris before and during production.
- Dispose of diseased plants and debris in a plastic trash bag. Keep the bag closed to help prevent spreading spores to uninfected plants as the bag is removed from the greenhouse. Cover trash cans to prevent the airborne spread of spores from diseased plant tissue.
- Reduce humidity and leaf wetness duration to prevent spore germination. Provide good air circulation and reduce humidity within the canopy.
- Proper planting dates, fertility, watering and height management will prevent overgrown plants, thereby reducing humidity within the canopy.
- Water in the morning, never late in the day. Rising temperatures during the day will cause water to evaporate from the foliage and dry the leaf surface.
- Avoid growing ornamental hanging baskets above vegetable bedding plants. Spent flowers dropping on plants below cause *Botrytis* infection.

Late Blight

Late blight is caused by the water mold *Phytophthora infestans*. The fungus typically overwinters in potato cull piles or in soil where plant tissue has not completely frozen and is not considered a problem for locally grown tomato seedlings.

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Symptoms: Common symptoms on tomatoes are sunken, dark green or brown lesions on leaves and brown lesions on stems, sometimes white fungal growth develops under moist conditions. Leaf lesions begin as irregularly shaped olive green to brown spots and quickly grow larger – spots that are consistently small are most likely *Septoria* leaf spot. Confirm Late Blight by submission of a sample to a diagnostic laboratory.

Management: Many of the same fungicides used for Botrytis blight will help protect tomato seedlings from late blight.

Damping-off of Seedlings

Damping-off is a common disease of germinating seeds and young seedlings. Several fungi are capable of causing damping-off including *Rhizoctonia, Alternaria, Sclerotinia* and the water molds, *Phytophthora* and *Pythium.* Soil-borne fungi generally do not produce air-borne spores but are easily transported from contaminated soil to pathogen-free soil by infected tools, hose ends, water-splash and hands. Young seedlings are most susceptible to damping-off. However, later in the crop cycle, the same pathogens may cause root and stem rot.

Symptoms: Symptoms of damping-off include seedlings failing to emerge or wilting, often with a stem lesion that appears water-soaked or dark, necrotic and sunken at the soil line. Pathogens usually spread radially from a central point of origin so plants often die in a circular pattern. Vegetable seeds that are germinated in poorly drained, cool soils are especially susceptible. Young plants that do emerge are weak and often wilt at or below the soil line. Cabbage, cauliflower, tomato and pepper seedlings may be girdled by brown or black sunken cankers. Stems of these plants may shrivel and become dark and woody (wirestem or collar rot). The plants may not collapse, but remain stunted and die after transplanting.

Management: Damping-off must be prevented because it is difficult to stop once symptoms occur. There are several strategies to prevent damping-off.

- Use only certified disease-free seed from reputable seed companies.
- Use fungicide-treated seed. Certain fungicides are labeled for damping-off for selected vegetable crops.
- Use pasteurized soil, properly produced compost-based or soilless mixes. Incorporate biological fungicides into your soilless mix or apply biological fungicides as a drench at planting.
- Disinfect all flats, cold frames, pots and tools.
- Germinate seed under conditions that will ensure rapid emergence, such as with the use of bottom heat.
- Avoid overwatering, excessive fertilization, and damaging roots during transplanting, handling, and planting too deeply.
- Fill flats with pre-moistened growing media to avoid compaction. Lightly fill and brush containers. Do not pack young plants into containers, use pre-dibbled holes for transplants.
- To avoid compaction, do not stack or “nest” filled trays or pots.
- Provide adequate light for rapid growth.
- Discard entire infected flats.

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Downy Mildew

Downy mildew (*Peronospora balbahrii*) on basil is a recent problem facing growers of basil (both in the greenhouse and in the field). It was first reported in Florida in 2007 and has been found in CT since 2008.

Symptoms: Infected leaves develop a diffuse yellowing that is easily confused with nutrient deficiency. Distinct vein bounded patches on the underside of the leaves develop that produce dark purple brown sporangia. The fuzzy, dark growth makes leaf undersides appear dirty.

Management: Management of environmental conditions such as temperature, humidity and duration of leaf wetness, sound cultural practices and fungicides will help prevent disease development.

- It is vital to reduce humidity and leaf wetness duration to prevent spore germination. See techniques for reducing relative humidity.
- Provide good air circulation and reduce humidity within the canopy. Proper planting dates, fertility, watering and height management will prevent overgrown plants, thereby reducing humidity within the canopy.
- Water in the morning, never late in the day. Rising temperatures during the day will cause water to evaporate from the foliage and dry the leaf surface.

Few fungicides are labeled for herbs grown in the greenhouse. See Pest Management for Herb Bedding Plants Grown in the Greenhouse and listed Tables referenced in the manual can be downloaded separately.

Powdery Mildew

Powdery mildew may occasionally occur on vegetable transplants including tomato, eggplant and other solanaceous crops, as well as cucurbit crops. Faint, white mycelium may develop on leaves and stems, with yellow margins.

Most growers are all too familiar with powdery mildew when it develops on cucurbits in the field. The powdery mildew that affects certain cultivars of ornamental verbena can also infect curcurbit seedlings including squash, cucumbers and pumpkins. Greenhouse growers who produce curcurbit transplants as well as verbenas should be especially careful to separate these two crops. It is possible that this powdery mildew could affect the cucurbit transplants that may not have otherwise become infected until the fruit was beginning to form in the field.

Bacterial diseases

Bacterial diseases of vegetable bedding plants, such as bacterial leaf spot of peppers and tomatoes, bacterial speck & bacterial canker of tomatoes, and black rot on cole crops are introduced into a greenhouse through infected seed and transplants.

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Bacterial leaf spot, Bacterial speck

Symptoms: Bacterial leaf spot is caused by *Xanthomonas campestris* pv. *vesicatoria* and is found primarily on peppers although all aboveground parts of tomatoes are also susceptible. Spots on leaves are chocolate-brown with yellowing at lesion's margins and irregularly shaped with areas of dead leaf tissue. At first, the spots are less than 1/4 of an inch in diameter. Severely spotted leaves will appear scorched and defoliation may occur. This disease is most prevalent during moderately high temperatures and long periods of leaf wetness.

Bacterial speck occurs on tomato but not pepper. The bacterium, *Pseudomonas syringae* pv. *tomato,* causes small black spots to develop resulting in chlorosis (yellowing), necrosis (dead tissue) and blighting of the foliage. Bacterial speck can usually be distinguished from bacterial spot by the size of the lesions, however, in some cases, the symptoms look similar.

Bacterial canker

Bacterial canker of tomato is caused by *Clavibacter michoiganensis* pv. *michiganensis* (formerly *Corynebacterium michiganense).* In New England, bacterial canker occurs less frequently than other tomato diseases but it is potentially more destructive. The bacterium is seed-borne but may survive on plant debris in soil for at least one year. It can also survive in the greenhouse on wooden stakes and flats. Wilt, leaf scorch, canker, pith necrosis and fruit spot may occur singly or in combination depending on the circumstances. When the bacterium is carried in the seed, the vascular system becomes colonized, resulting in wilt, pith necrosis and external cankers. Wilt initially occurs on one side of a leaf or one half of a plant because only a portion of the vascular system is blocked. Cankers and pith necrosis occur in later stages of disease development. Cankers are dark and water-soaked in appearance and often exude bacteria that are easily spread to adjacent plants. Pith necrosis is first evident as a darkening of the center of the stem that soon becomes chambered or hollow. When leaf scorch occurs, the petioles usually bend downward while the leaf edges curl up. The margins of the leaves become brown with a yellow border to the inside. Scorching of the foliage often develops in the absence of wilt or stem canker. Transplants may not express symptoms until six to eight weeks after infection and initial symptom expression is accelerated by environmental stress.

Black rot

Black rot, caused by the bacterium *Xanthomonas campestris* pv. *campestris* occurs where cruciferous plants are grown. All Brassicas can be severely affected. The bacterium enters the leaves by colonizing the hydathodes (water pores) and moves from the leaf margins inward. Lesions may also begin at wounds. Diseased tissue is often V-shaped; flaccid, tan to yellow and with blackened veins. The blackened veins are diagnostic and are best seen by holding the leaf up to the light. When the lesions reach the petiole and stem, the bacterium moves systemically through the plant, resulting in premature leaf drop. At this stage of disease, a cross-section of the stem will reveal a ring of discolored vascular tissue.

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Management of bacterial diseases: These bacteria can be introduced on infected seeds, infected transplants purchased from another operation, or in the field on crop residues. These bacteria can also survive on weeds in the same family as the host crop. The management of these bacterial diseases is similar and includes the following strategies:

- Buy certified disease-free seed from a reputable source.
- Use hot water-treated seed. Ideally, the seed should be custom-treated by the seed company. Seed companies may treat the seed upon request. There is a risk that germination percentages will be reduced if the seed crop is grown under stressful environmental conditions.
- Promptly remove infected plants and adjacent plants to prevent further infection and avoid unnecessary handling of plant material.
- Avoid overhead irrigation, splashing or periods of extended leaf wetness.
- Disinfect all benches, equipment, flats and stakes.
- Follow sound practices for weed and insect control.
- Prevent bacterial leaf spot on peppers by choosing resistant varieties whenever possible. There are many resistant varieties of bell peppers available, but few resistant specialty peppers.

Viral Diseases

Some viral diseases of vegetable bedding plants include cucumber mosaic virus (CMV), tobacco mosaic virus (TMV) and the tospoviruses, INSV and TSWV. There is no control for plants infected with a virus. It is important to have the virus disease accurately identified. Serological techniques such as ELISA (enzyme-linked immunosorbent assay) are now available to accurately identify a wide range of viruses. On-site grower kits using this same technology are also available from Agdia to test for viruses such as CMV, TMV, INSV, and TSWV.

Cucumber mosaic virus

Cucumber mosaic virus (CMV) has a wide host range of over 400 species of plants including vegetables, ornamentals and weed hosts.

Symptoms: Infected plants may show mild mosaic patterns and mottling, flecking, and fern leaf distortion.
CMV is primarily spread by aphids that can acquire the virus in as little as 5 to 10 seconds. Aphids then move the virus from plant to plant for a few hours.

Management: Rogue diseased plants. Eliminate weeds such as common pokeweed, chickweed, field bindweed, yellow rocket, and bittersweet nightshade that may be reservoirs of CMV.

Tobacco Mosaic Virus (TMV)

TMV has a wide host range but is especially a concern on solanaceous crops. In recent years, TMV has been reported on pepper, petunia, and tomato. TMV is not transmitted by insects! It is a very stable virus that can be spread by contact. Workers can easily spread TMV when they handle plants or when cutting tools become contaminated. TMV can persist in dried tobacco leaves, so tobacco products can also be a source of TMV.

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Symptoms: Symptoms include yellow mottling, upward leaf curling and overall stunting. Some infected plants may not show any symptoms at all.

Management: Discard infected plants including roots. Disinfect hands by washing with milk, or tri-sodium phosphate and then thoroughly with soap and water. Smokers need to wash their hands before entering the greenhouse so they do not infect plants. In greenhouses, hard surfaces such as doorknobs, or flats can become contaminated after handling virus-infected plants and remain a source of infection. Thoroughly disinfect the growing area with a commercial disinfectant. Control perennial weeds in the solanaceous family such as ground cherry and horsenettle that could be reservoirs of TMV.

Tospoviruses

Tospoviruses are a group of viruses that include impatiens necrotic spot virus (INSV) and tomato spotted wilt virus (TSWV). They may infect hundreds of plant species including basil, tomatoes, peppers and eggplant. These viruses are primarily spread by the western flower thrips. Tospoviruses are not seedborne but are brought into the greenhouse on vegetatively propagated ornamental plants or seedlings that have been exposed to the virus. Once the thrips in the greenhouse become infected, they can transmit the virus to susceptible crops and weeds.

Symptoms: Symptoms include stunting, foliar ringspots and black lesions on stems. Symptoms of INSV and TSWV will vary depending upon the host.

Management: To manage Tospoviruses, it is necessary to discard infected plant material, including weeds and to manage thrips. Infected vegetable transplants planted into the garden or field will be stunted and will not produce a harvestable crop. Since INSV and TSWV are not seed-borne, vegetable transplants may be kept free of Tospoviruses if they are not brought into contact with other infested crops or thrips carrying the virus. Growers attempting to concentrate all their warm temperature crops in a single house run a risk of mixing Tospovirus-free vegetable seed crops with leftover ornamental stock plants or new cuttings that may carry the virus. Pre-finished or vegetatively propagated ornamentals from another producer could be infested with thrips or a virus. Therefore, vegetable bedding plants should always be grown in separate greenhouses.

General Insect and Mite Pest Management

Monitoring

A regular monitoring program is the basis of all pest management programs. Conduct a regular, weekly scouting program to detect pest problems early. This early detection and treatment will result in better pest control since plant canopies are smaller and better spray coverage can be achieved.

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Yellow Sticky Cards

Use yellow sticky cards to trap and detect adult stages of flying insects such as western flower thrips (WFT), whiteflies, fungus gnats, shoreflies, leafminers and winged aphids. Remember that mites and wingless aphids do not fly and will not be caught on the cards. Place one to four cards per 1,000 square feet. The cards should be spaced equally throughout the greenhouse in a grid

pattern with additional cards located near doorways and vents. Place some cards just above the plant canopy (to detect thrips and whiteflies) and some of the cards on the rim of the flats or pots to detect fungus gnats. Inspect and replace the cards weekly to keep track of population trends.

Plant Inspection

Plant inspection is needed to assess general plant health and to detect diseases, mites and wingless aphids plus any hot spots of immature whiteflies. Randomly select plants at ten locations in an area of 1,000 square feet, examining plants on each side of the aisle. Start this pattern at a slightly different location each week, walking through the greenhouse in a zigzag pattern down the walkway. Examine the underside of leaves for insect pests and inspect root systems to determine whether they are healthy.

Key Plants and Indicator Plants

Focus on scouting key plants and indicator plants. Key plants are those plants or cultivars that have serious, persistent problems every year. For example, peppers and eggplants are prone to aphid infestations. Look for aphids on the young leaves and for shiny honeydew on the upper leaf surface. If grown near flowering plants, peppers and eggplant will also indicate an early thrips population. Look for distorted, young leaves with silvery flecked scars that are signs of thrips feeding damage.

Fava beans and certain cultivars of petunia are used as indicator plants to detect the presence of thrips carrying INSV and TSWV. These plants will develop viral symptoms within one week if fed on by the infected thrips. The petunia cultivar 'Summer Madness' and several varieties of fava bean have been successfully used to detect tospoviruses. To use petunias and fava beans as indicator plants:

- Remove flowers from indicator plants to encourage feeding on foliage where symptoms can be observed.
- Place a blue non-sticky card in each pot at plant height. The blue card will attract thrips to the indicator plant. Blue plastic picnic plates work well.
- Place petunia plants throughout the greenhouse among the crop at a rate of one plant every 20-30 feet and fava bean plants at the rate of 12 pots per 1,000 sq. ft.
- Remove symptomatic leaves on petunia plants and continue to use the plants. The virus is not systemic in these plants. Thrips feeding injury leaves distinct white feeding scars on the foliage. Virus symptoms appear as a brown rim around the feeding scars.
- Remove entire plants of fava beans if symptoms are observed, because the virus is systemic in these plants. Viral symptoms appear as dark brown angular lesions on leaves or yellow to light green ring spots. Dark necrotic areas can also be seen on the stem. Fava beans have dark black spots on their stipules that should not be confused with viral symptoms.
- Replace with new plants, planting 1-2 bean seeds per 4" or 6 "pot.

Record Keeping and Decision-Making

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Each time the crop is scouted, record the pest numbers, their location and the number of plants inspected. Records on pest numbers and locations will help you identify population trends. Population trends will also indicate if initial control measures were successful or if they need to be repeated. Once this information is collected each week, a pest management decision can then be made. Monitoring and record keeping will answer the following questions and help you make the necessary treatment decisions. Is the population decreasing, increasing or remaining stable over the growing season? Do you need to spray? Are insects and mites migrating from weeds under the benches to your crops? Is the treatment from last week working? Tables 1 and 2 provide a list of selected materials labeled for managing insects, mites and diseases on greenhouse-grown vegetable transplants. Follow label instructions before using the material on vegetable bedding plants. The product must be used only for crops for which the compound is registered.

Biological Control for Insects and Mites

Biological control may be an option for aphids, mites, fungus gnats, thrips and whiteflies. Natural enemies are living organisms that need to be released when pest populations are low. They do not act as quickly as pesticides so cannot be used as a “rescue” treatment. Natural enemies (parasites, predators or pathogens) are best used early in the cropping cycle when plants are small, pest numbers are low and damage is not yet observed. A detailed plan of action is needed to insure success. Accurately identify the key pests in your production system. Natural enemies, especially parasites, are often very specific to a particular pest. Many insecticide residues can adversely affect natural enemies for up to 3 months after their application. Koppert Biological Systems has compiled a list of insecticides and their effects on natural enemies. This list is available from Koppert Biological Systems , Inc., 2856 South Main St., Ann Arbor, Michigan 48103. Biobest Biological Systems also has a searchable pesticide side effects database.

When using beneficial nematodes, refer to the online database compiled by Becker Underwood (check under Chemical Compatibility Guide). Become familiar with using insecticides and miticides that are compatible with natural enemies such as insecticidal soap, horticultural oil and certain insect growth regulators and neem-based materials (azadirachtin products)**,** see Table 1 Selected Insecticides Labeled for Insects and Mites on Vegetable Bedding Plants and have a sprayer dedicated for their use.

Start in a small trial area to become familiar with releasing, monitoring and evaluating the effectiveness of natural enemies. A separate greenhouse is best. With help from your biological control agent supplier, establish a schedule for introducing the natural enemies. Release rates and timing will vary depending upon the crop and its size, the degree of infestation, effectiveness and type of natural enemies, plus the time of year. Starting a biological control program will involve some trial and error as release rates have not been scientifically evaluated for vegetable bedding plants. Vegetable bedding plants with only one or two key insect pests or with a longer production schedule may be logical candidates for biological control. Be sure that natural enemies are received from your supplier quickly (4 days), and that they are not exposed to temperature extremes (too hot or cool) during shipment. Request that the biological control supplier include ice packs and a data logger (if possible). Inspect natural enemies for viability and

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quality when they are received. Ask your biological control supplier how to best evaluate any incoming shipments. Table 4 Scouting Guidelines and Biological Control Options for Vegetable Bedding Plants and Transplants provide information on scouting for key insect and mite pests and biological control options.

Specific Insect Pests and Mites

Common insect pests on vegetable bedding plants include aphids, fungus gnats, shore flies, whiteflies, thrips and two-spotted spider mites. The following are brief descriptions, life cycles and monitoring tips for the major pests. See Table 4 Scouting Guidelines and Biological Control Options for Vegetable Bedding Plants and Tables 1 Selected Insecticides Labeled for Insects and Mites on Vegetable Bedding Plants additional scouting guidelines, registered pesticides and biological control options.

Aphids

Lifecycle: Several species of aphids can occur on vegetable transplants, but the most common are green peach, melon and foxglove. Aphids are small, 1/16-inch in length, round, soft-bodied insects that vary in color from light green to pink or black. The green peach aphid is yellowish-green in summer; pink or yellowish in fall and spring. Winged forms are brown with a large dusky blotch on the abdomen. Melon aphids are greenish-yellow to very dark green with black mottling and short dark cornicles or “tailpipes” (tubular structures on the posterior part of the abdomen). Foxglove aphids are smaller than potato aphids but larger than melon and green peach aphids. The foxglove aphid is a shiny light yellowish green to dark green in color with a pear-shaped body. The only markings on the bodies of wingless adults are dark green patches at the base of the cornicle. The legs and antennae also have black markings. Foxglove aphids cause more leaf distortion than green peach or melon aphids. Aphids feed by inserting their piercing, sucking mouthparts into plant tissue and removing fluids. In greenhouses, aphids are usually females that produce live young called nymphs. Each female can produce 50 or more nymphs. Nymphs mature to adulthood and begin reproducing in as little as 7 to 10 days. Adults are usually wingless, but some will produce wings when populations reach outbreak levels. Large numbers of aphids will stunt and deform plants. In addition, aphids produce a sticky digestive by-product called honeydew and their white shed cast skins may be unsightly. Sometimes, these white cast skins are mistakenly identified as whiteflies. Honeydew can cover leaves and provide a food source for a superficial black fungus known as sooty mold. Aphids are present on weeds and winged aphids may enter the greenhouse through vents. Aphids can also transmit certain viruses.

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Monitoring: Examine the foliage, along stems and new growth of key plants such as peppers, eggplants, cole crops and leafy greens to detect an early aphid infestation. Signs of aphid activity include shed white skins, shiny honeydew, curled new leaves, distorted growth and the presence of aphids. Yellow sticky cards help detect the entrance of winged aphids into the greenhouse from outdoors. Yellow cards will not, however, allow you to monitor aphids within the crop, as most of the aphids will be wingless.

Whiteflies

Lifecycle: The sweetpotato whitefly *(Bemisia argentifolii)* and greenhouse whitefly *(Trialeurodes vaporariorum)* may infest vegetable bedding plants. However, greenhouse whitefly is the most common species. Both adult and immature whiteflies have piercing sucking mouthparts, are able to remove fluids and produce honeydew that also results in sooty mold fungus. Winged adult whiteflies are 1/16-inch in length, and found on the undersides of the youngest, most tender leaves. Females may lay from 150 to 300 eggs, which hatch into first-instar nymphs in about a week. The crawlers move for a short distance before settling down to feed. After three molts, a pupal stage is formed from which adults emerge in about six days. Whiteflies complete their egg to adult cycle in 21 to 36 days depending upon greenhouse temperatures.

Monitoring: To monitor whiteflies, check susceptible plants, such as tomatoes, at ten locations in an area of 1,000 square feet, examining plants on each side of the aisle. Look on the undersides of one or two leaves per plant, for nymphs, pupa and adults. Yellow sticky traps can also be used to detect adult whiteflies once populations have reached higher densities. Begin treatments as soon as the first sign of infestation is noted.

Fungus Gnats, Shore Flies and Predatory or Beneficial Hunter Flies

Lifecycle: The damp, moist environment in greenhouses favors both fungus gnats and shore flies. Fungus gnat larvae are translucent, white and legless, about 1/4 inch long when mature, and have a shiny black head. The mosquito-like adult is about 1/8 inch long, with long legs, a pair of clear wings and long antennae. There is a distinct “Y” vein on each wing. Fungus gnats are weak fliers and are frequently observed resting on pot media or running over the foliage or other surfaces. The larvae feed on fungi and decaying organic matter, and often injure seedlings and plants. Larva feeding occurs on young, tender roots and in the stem at the base of the plant. This feeding injury provides an entry for disease pathogens. A female fungus gnat may lay up to 300 whitish eggs in clusters of 20 or more. The eggs are deposited on the surface or in the crevices of moist soil or potting media. Eggs hatch in about six days. Larvae feed for 12 to 14 days before changing into pupae. The pupal stage may last five to six days. Adults live up to ten days. The life cycle from egg to adult requires approximately four weeks depending on greenhouse temperatures.

Adult shore flies also occur in damp greenhouses. Shoreflies are often misidentified as fungus gnats or hunters flies but have a distinctly different appearance. The adult shore fly is about 1/8 inch long and has a robust body, very short antennae, shorter legs and dark wings with about five light spots. Adults may be seen resting on plant leaves. Larvae are off-white and do not have distinct head capsules that are characteristic of fungus gnat larvae. Shore flies do not injure plants through direct feeding, but can carry root rot pathogens from diseased to healthy plants. Their fecal spots or droppings can also be unsightly. To manage shore flies, control their food source, algae.

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Adult hunter flies, a beneficial predatory (beneficial fly) are also found on sticky cards that may be mistaken for shore flies. Hunter flies can be distinguished from shore flies, by their size and color. Hunter flies are larger, gray in color (males are lighter gray than the females) and do not have

light spots on their wings. Hunter flies are in the same family as common houseflies and are similar in appearance, but smaller. Hunter flies may prey upon fungus gnats, shore flies, *Liriomyza* leafminers, moth flies and whiteflies.

Monitoring: To monitor for fungus gnat larvae, place raw potato chunks (with peel removed) on the soil surface. Larvae are attracted to the potato chunks and will congregate underneath. Check the potato chunks after 2 days for the larvae. Potato disks cut one inch in diameter and 0.5-1 inch thick are effective. In addition, choose plants on each bench and inspect the soil surface and around the base of the plant including the stem just below the soil line. Record the location and the level of infestation. Badly infested plants should be removed as they serve as a source of infestation.

Adult fungus gnats can be monitored with yellow sticky cards placed at the base of the plant at the soil line. Weekly inspections of yellow sticky cards can detect the onset of an infestation, and continued recording of the number of adults per card per week can aid in evaluating the efficacy of control efforts.

Thrips

Lifecycle: The most injurious species is the western flower thrips (WFT). They often do considerable damage before they are discovered, because thrips are small, multiply rapidly and feed in plant buds in which they can remain undetected. WFT also vector tospoviruses. Feeding marks from the rasping mouthparts of thrips appear as white streaks on the leaves. Infested new growth may curl under and leaves are often deformed. Adult WFT are about 1/16-inch long, with narrow bodies and fringed wings. Females are reddish brown and males are light tan to yellow. The wingless immature larval stages are light yellow. Female thrips insert eggs (several hundred per female) into plant tissue. The tiny yellowish larvae molt twice and feed on plant fluids as they mature. Larvae drop off the plant into the soil and pass through two stages, after which adults emerge. The egg to adult lifecycle can be completed in 7 to 13 days depending upon greenhouse temperature. During warmer temperatures development is more rapid than at cooler temperatures.

Monitoring: Early detection of a thrips infestation is critical for effective management because populations are lower and it is easier to obtain good spray coverage when plant canopies are small. Symptoms of their feeding are often not noticed until the damage has occurred. Eggplant, tomatoes, and peppers are prone to thrips infestations. Yellow sticky cards, key plants and indicator plants provide an easy way to detect the onset of an infestation. Yellow sticky cards should be placed just above the crop canopy, and near doors, vents and over thrips sensitive cultivars to monitor the movement of thrips. The light to medium-blue sticky cards may catch more thrips (and shore flies) than yellow ones. However, it is more practical to use yellow cards for general pest monitoring to attract fungus gnats, whiteflies and winged aphids. The number of thrips per card should be recorded weekly and graphed weekly to monitor population levels and movement in or out of the greenhouse, and thus aid in control decisions. See section on key plants and indicator plants for more monitoring information.

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Spider Mites

Lifecycle: Two-spotted spider mites can be found on vegetable bedding plants. Adult females are approximately 1/50-inch long, and slightly orange in color. All mobile stages are able to pierce plant tissue with their mouthparts and remove plant fluids. Most spider mites are found on the underside of leaves. Feeding injury often gives the top leaf surfaces a mottled or speckled, dull appearance. Leaves then turn yellow and drop. Large populations produce visible webbing that can completely cover the leaves. Eggs are laid singly, up to 100 per female, during her 3 to 4-week life span. Eggs hatch into larvae in as few as 3 days. Following a brief larval stage, several nymphal stages occur before adults appear. Egg to adult cycle can be completed in 7-14 days depending upon temperature. Hot and dry conditions favor spider mite development.

Monitoring: Check for mites by examining foliage. Adult spider mites are not found on sticky cards. Mites often develop as localized infestations on beans, tomatoes, or eggplants. Sample plants by turning over leaves and with a hands-free magnifier (Optivisor™) or hand lens, check for the presence of spider mites.

Cyclamen Mites

Life Cycle: The shiny, orange-tinted cyclamen mites prefer to hide in buds or deep within the flowers. Adult females can lay from 2 to 3 eggs per day for up to two to three weeks. Eggs are deposited in moist places at the base of the plant. Cyclamen mites can complete their life cycle in 1 to 3 weeks. Females can live up to one month and can reproduce without mating. Cyclamen mite females lay 2 to 3 eggs per day for up to two to three weeks. Cyclamen mite eggs are oval, smooth and about one half the size of the adult female. Larvae hatch from the eggs in 3 to 7 days. The slow moving white larvae feed for 4 to 7 days. Cyclamen mites prefer high relative humidity and temperatures of 60o F. Cyclamen mites affect a number of ornamental bedding plants including dahlia, fuchsia, gerbera daisy, petunias, viola as well as strawberries in the field. They may migrate to peppers or tomatoes.

Monitoring: Cyclamen mites pierce tissue with their mouthparts and suck out cell contents. Look for signs of damage which may be concentrated near the buds or occur on the entire plant. Symptoms include inward curling of the leaves, puckering and crinkling. Pitlike depressions may develop. The mite is only 1/100 of an inch long. Examination under a microscope is often needed to confirm the presence of cyclamen mites.

Broad Mites

Life Cycle: Broad mites are closely related to cyclamen mites. They can be distinguished from cyclamen mites by their egg stage. Eggs are covered with “bumps” that look like a row of diamonds. Eggs are best seen using a dissecting microscope. Adults and larvae are smaller than the cyclamen mites and walk rapidly on the underside of leaves. Broad mites can also be attach themselves to whiteflies and use the whiteflies as a carrier for their dispersal. The development of broad mites is favored by high temperatures (70 to 80o F). Broad mites can complete their life cycle in as little as one week. Females lay from 30 to 75 eggs.

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Monitoring: Broad mites can affect a number of ornamentals including gerbera daisy, New Guinea Impatiens, saliva, ivy, verbena and zinnia. They may migrate to peppers or tomatoes. Look for characteristic damage including leaf edges curling downward. Terminal buds may be killed. As they feed, broad mites inject toxic saliva, which results in the characteristic twisted, distorted growth. Broad mite injury can be mistaken for herbicide injury, nutritional (boron) deficiencies or physiological disorders**.** Inspect the underside of the leaves for the mites and their eggs with a 20x hand lens or submit samples to a laboratory for diagnosis. Microscopic examinations are often helpful.

Weed Management

In greenhouses, weeds are a primary source of insects such as aphids, whiteflies, thrips, and other pests such as mites, slugs and diseases. Low growing weeds help maintain moist conditions, a favorable environment for fungus gnats and shore flies. Many common greenhouse weeds such as chickweed, oxalis, bittercress, jewelweed, dandelion and ground ivy can become infected with tospoviruses including impatiens necrotic spot virus (INSV) and tomato spotted wilt virus (TSWV) while showing few, if any visible symptoms. Thrips can then vector the virus to susceptible greenhouse crops. Weeds can also carry other plant damaging viruses that are vectored by aphids. An integrated weed management program will help to effectively manage weed populations. This approach includes preventive measures such as sanitation and physical barriers, and control measures such as hand weeding and the selective use of postemergence herbicides.

The use of a physical barrier such as a weed block fabric is an effective method to limit weed establishment on greenhouse floors. The weed fabric should be left bare so it can be easily swept. Covering the weed fabric with gravel makes it difficult to remove any spilled potting media providing an ideal environment for weed growth. Regularly pull any escaped weeds before they go to seed. Repair tears in the weed block fabric.

Few herbicides are labeled for use in a greenhouse due to the potential for severe crop injury or death to desirable plants. This injury may occur in a number of ways including: 1) spray drift if fans are operating at the time of application, and 2) volatilization (changing from a liquid to a gas). Herbicide vapors are then easily trapped within an enclosed greenhouse and injure desirable plant foliage. Always be sure the herbicide selected is labeled for use in the greenhouse. Carefully follow all label instructions and precautions. It is the applicator's responsibility to read and follow all label directions. Use a dedicated sprayer that is clearly labeled for herbicide use *only.*

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Avoid use of preemergence herbicides in the greenhouse! Preemergence herbicides are applied to soil to prevent the emergence of seedlings. There are currently no preemergence herbicides labeled for greenhouse use. They can persist for many months and in some cases over a year. Preemergence herbicides can volatilize or vaporize in the greenhouse, causing significant damage to young transplants. (Note: Surflan (oryzalin) is no longer registered for use in enclosed greenhouses).

Postemergence herbicides are applied after the weeds have emerged. Several postemergence herbicides can be used under greenhouse benches and on the floors. Contact herbicides are best applied to small seedlings. Large weeds will be burned but not killed.

Herbicides for use in greenhouses

Clethodim (Envoy Plus)*:* Selective, postemergence herbicide, for the control of grasses only, works by contact. For use when crops are in the greenhouse.

Clove oil (Matratec): Non-selective, postemergence herbicide. Works by contact. For use when crops are in the greenhouse.

Diquat dibromide (Diquat E Pro ZL, Reward)*:* Non-selective. Works by contact. For use when crops are in the greenhouse.

Glyphosate (Glyphosate Pro 4, Razor, Roundup Pro, Roundup Pro Concentrate, Touchdown Pro): Non-selective postemergence herbicides. Systemic. For use in an empty greenhouse between crops and outside greenhouses.

Pelargonic acid (Scythe): Non-selective, postemergence, contact herbicide. Cool or cloudy weather may slow down activity. Provides no residual weed control but leaves a strong odor. For use when crops are in the greenhouse.

Rosemary oil, clove oil, thyme oil (Sporatec): Non-selective, postemergence herbicide. Works by contact. For use when crops are in the greenhouse.

Weed control outside greenhouses

In addition to mowing, herbicides may also be used outside of greenhouses. Before spraying weeds around the greenhouse with any herbicide, close windows and vents to prevent spray drift from entering the greenhouse. Avoid using auxin-type herbicides, such as those labeled for broadleaf weed control in turf or brush killers, or herbicides with high volatility near greenhouses. Select herbicides with low volatility.

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New England Greenhouse Update – This website consists of both greenhouse updates and photo gallery. The photo gallery provides photos and descriptions of hundreds of plant problems caused by insects, mites, diseases, nutritional disorders and cultural procedures.

University of Massachusetts Extension Greenhouse Crops and Floriculture Program

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Helpful Websites on Organic Production
Biernbaum, John. 2006. Greenhouse Organic Transplant Production .pdf □ .pdf . Michigan State University .pdf □ .pdf

Publications:

Organic Greenhouse Vegetable Production, Potting Mixes for Certified Organic Production, Organic Greenhouse Tomato Production, Plug and Transplant Production for Organic Systems
ATTRA – National Sustainable Agriculture Information Service □ □

The National Organic Program Guidelines □

□

Cornell University Agricultural Experiment Station
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Helpful Information on Greenhouse Management/Engineering

Aldrich, R. A. and J. W. Bartok. 1994. Greenhouse Engineering, NRAES-33. 199 pp. Available from NRAES (Natural Resource, Agriculture and Engineering Service □)

Factsheets on Greenhouse Management and Engineering □

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